

ICST INETRDISCIPLINARY CENTER FOR SMART TECHNOLOGIES

Center of CS Department's Software Engineering Track

SmartRail

Know where to board.

Problem: Passengers lack seat information before boarding.

SmartRail: An IoT system for seat-aware platform guidance.

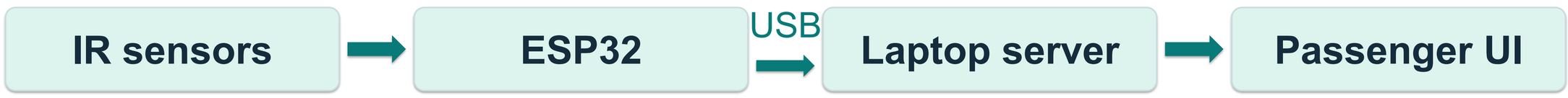
- **Seat sensing:** ESP32 and IR sensors detect occupied seats.
- **Hybrid train:** 12 x 108 seats (1,296); two physical inputs.
- **Web guidance:** Car counts, seat maps and zone suggestions.

Method: Calibration, filtering and USB state updates.

Advantage: Check before boarding, not after entering a car.

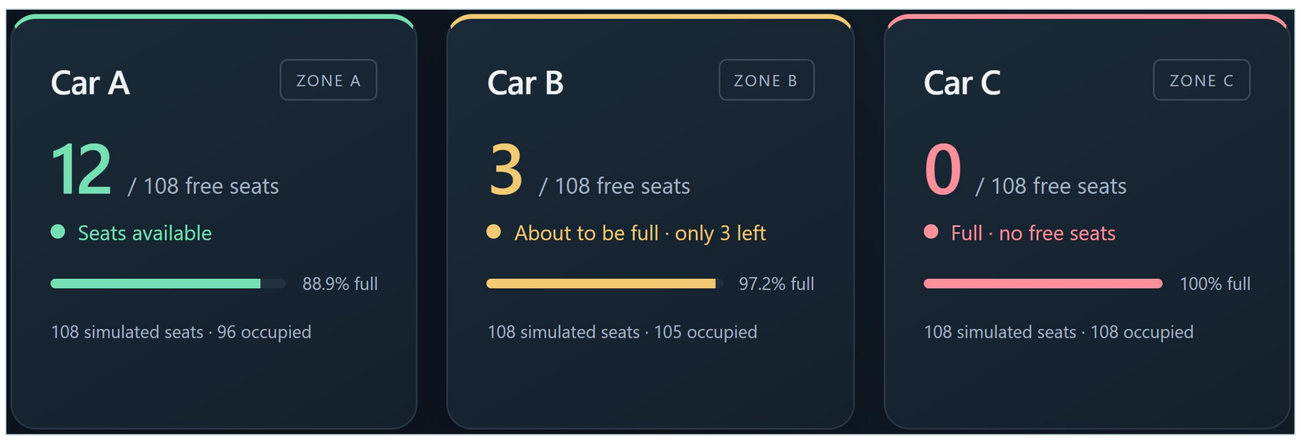
Demonstration: 2 physical seats + 1,294 simulated seats.

Conclusion: One display combines real and simulated seats.

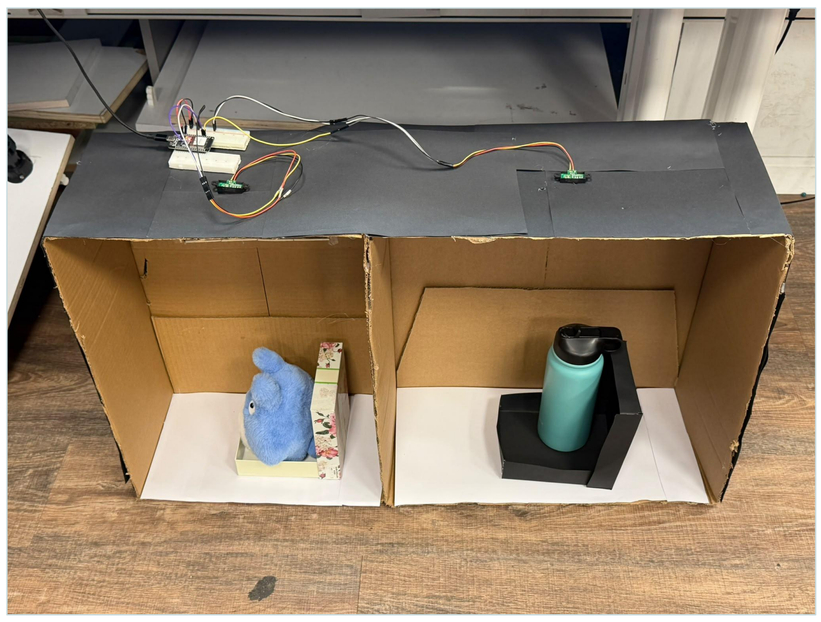




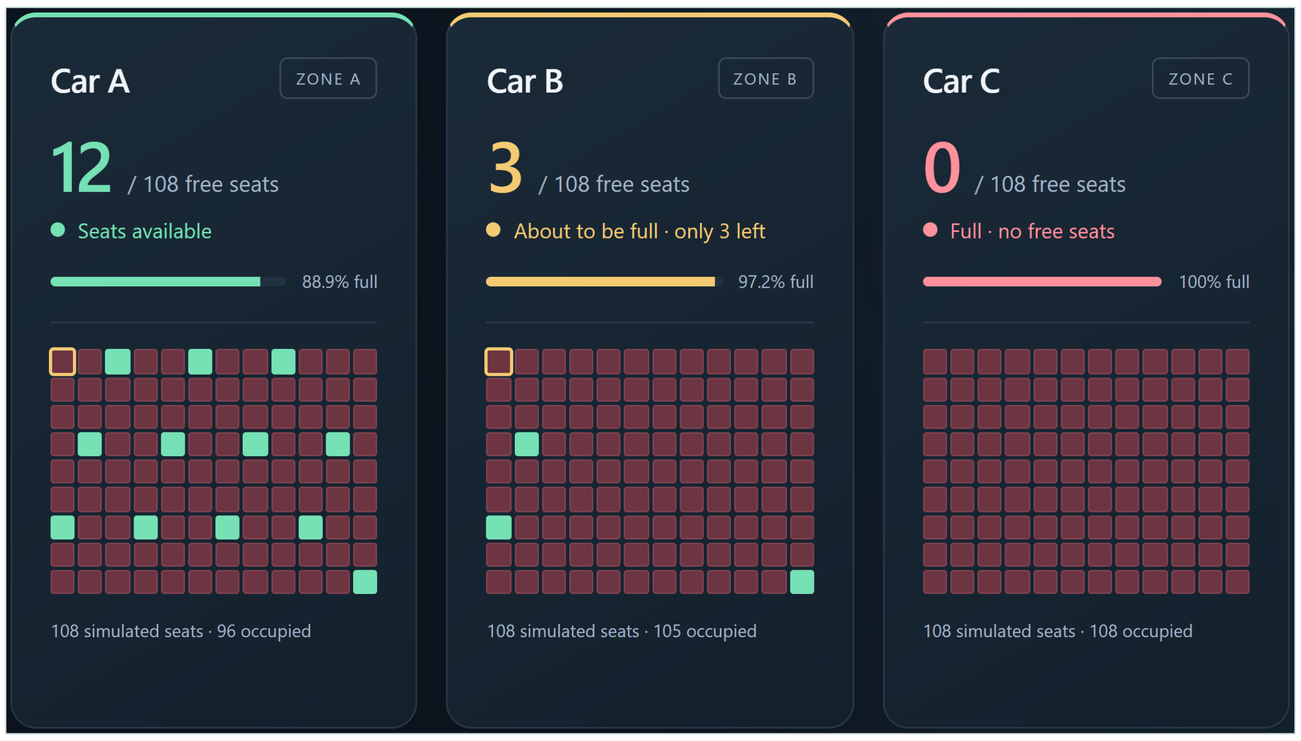
Integrated hardware + web demo



Passenger availability display — simulation excerpt



Two IR-sensed seat models



Operator seat map — simulation excerpt (A–C)

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github.com/abedazem13/SmartRail

A Project in Internet of Things (IoT)

Computer Science Department

הטכניון מכון טכנולוגי לישראל